Arya Vaidya

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Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Science. Threads: AI and Devices May 2029
- Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, Linear Algebra

Bridgewater-Raritan High School Sep 2022 - Jun 2026
GPA: 3.97, Top 5% of Class
- Relevant Coursework: Calculus III, Differential Equations, AP Probability & Statistics (5)

Skills

Languages & Tools: Python, Java, C++, SQL, Git, Google Firebase, Claude Code, Codex
Frameworks: PyTorch, OpenCV, Hugging Face, Jupyter Notebook, Stripe
App/Website Development: Swift, Xcode, React, Next.js, HTML, Tailwind CSS

Experience

MoBoo June 2025 – Sept 2025
Applied AI Engineer Intern *Remote*
- Generated consolidated book summaries across 1M+ tokenized reviews, by fine-tuning a Llama 3.2 LLM with LoRA-based parameter-efficient training via Hugging Face
- Ensured generated summaries scored 90%+ average faithfulness across 35k+ books, by designing an automated evaluation pipeline using a larger Llama 3.2 model as an LLM-judge
- Enabled 32k users to make more informed reading decisions, by building Python tools for BERT-based sentiment analysis on critic reviews and keyword extraction with KeyBERT

iCode Jan 2024 – Jul 2026
Instructor and Course Developer *Bridgewater, NJ*
- Taught 35+ students through project-based instruction in ML, Python, Arduino, and VEX Robotics, guiding them through hands-on capstone projects and competition-ready builds
- Built and sold a comprehensive 50-week AI curriculum to iCode covering ML, deep learning, and agentic AI, supported by original slide decks, quizzes, and hands-on Colab notebooks

Programming Projects

ML Hypernetwork Research for Fluid Simulations June 2025 – Present
- Engineered a lightweight hypernetwork architecture in PyTorch for multi-step fluid dynamic simulations across 4 datasets, applied transfer learning to extend models trained on 1D simulations to 2D, requiring 20× less training data
- Cut 50-step rollout MSE by 72.3%, by developing a calibrated periodic boundary-condition loss with warmup scheduling
- Co-authoring an AAAI paper on this work with Prof. Nikhil Muralidhar at the Stevens Institute of Technology

AlumHub Apr 2025 – Present
- Built a full-stack mentorship platform using Next.js and React to connect 3k+ high school students and 2k+ alumni, backed by Firebase for authentication and SQL for data storage
- Scaled the platform to 170+ schools across 24 states, by optimizing SQL queries 3x faster through database indexing and caching, and integrating Stripe for subscription payments
- Engineered a rule-based scoring algorithm to connect students and mentors on shared interests and experiences, with an LLM re-ranking layer for the top matches

Programming Supervisor and Subcaptain for FRC Robotics Sept 2022 – June 2026
- Managed a $50k+ budget and led 50 members to develop competition-ready robots and host 6 STEAM events
- Enabled autonomous navigation and scoring under pressure, by building a real-time YOLOv8 object detection pipeline that fed live game piece positions into an A* pathing algorithm
- Built a scouting app for 100+ members to log in-match robot performance data, syncing entries offline over peer-to-peer Bluetooth to a master device, enabling the scouting lead to make data-driven alliance selection picks
- Named a FIRST Leadership Award Semi-Finalist in March 2025

Generative AI Research for Art Restoration Dec 2024 – July 2025
- Designed a latent diffusion model for artwork restoration in PyTorch combining a residual VAE with an image-conditioned diffusion transformer, achieving 0.010 L1 error on artwork damaged by random patch masking
- Outperformed 4 state-of-the-art restoration models by 8.1% in PSNR and 1.82% in SSIM, by validating the architecture through ablation studies against 3 baseline architectures and 6 loss function configurations
- Authored a research paper and presented the work at the NJ ISEF Fair, placing 3rd in the Computer Science category